b. Find the dominant Eigen value and the corresponding Eigen vector for the following matrix
$$A=\begin{pmatrix}1 & 6 & 1\\ 1 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}$$

29. a.i From the following data find the number of students whose weight is between 60 and 70lbs.

Weight in lbs:	0-40	40-60	60-80	80-100	100-120
No. of students	250	120	100	70	50

ii. Express $f(x)=x^3-3x^2+5x+7$ in terms of factorial polynomial taking h=2 and find its differences.

(OR)

b. Find f(x) as a polynomial in x for the following data and also find f(3) by Newton's divided difference.

x:	–4	–1	0	2	5
f(x):	1245	33	5	9	1335

30. a. The population of a certain town is given below. Find the rate of growth of the population in 1931, 1941, 1961 and 1971.

Year x:	1931	1941	1951	1961	1971
Population in thousands y:	40.62	60.80	79.95	103.56	132.65

(OR)

b. The velocity v of a particle at a distance S from a point on its path is given by the table below:

S in meter:	0	10	20	30	40	50	60
V m/sec:	47	58	64	65	61	52	38

Estimate the time taken to travel 60 metres by Simpson's rules.

31. a. Using R.K method of 4th order, solve $\frac{dy}{dx}=\frac{y^2-x^2}{y^2+x^2}$ with $y(0)=1$ find the value of y at x= 0.2.

(OR)

b. Solve $y'=x-y^2, 0\le x\le 1, y(0)=0, y(0.2)=0.02, y(0.4)=0.0795,\ y(0.6)=0.1762$ by Milne's method to find y(0.8).

32. a. Solve $\nabla^2u=-10\left(x^2+y^2+10\right)$ over the square mesh with sides $x=0, y=0, x=3$ and $y=3$ with boundary u = 0 and mesh length 1 unit.

(OR)

b. Solve $\frac{\partial u}{\partial t}=\frac{\partial^2 u}{\partial x^2}$ in $0<x<5, t\ge 0$ given that $u(x,0)=20, u(0,t)=0, u(5,t)=100$. Compute u for one time – step with h = k = 1 by Crank – Nicholson method.

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Reg. No.															

B.Tech. DEGREE EXAMINATION, MAY 2017
Third / Fourth Semester

15MA206 – NUMERICAL METHODS
(For the candidates admitted during the academic year 2015 – 2016 onwards)

Note:
(i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours Max. Marks: 100

PART – A (20 × 1 = 20 Marks)
Answer **ALL** Questions

1. The principle of method of least squares states that the best fitting curves to the set of points is that for which the sum of squares of residual is
(A) Minimum (B) Maximum
(C) Greater than zero (D) Less than zero

2. Newton-Raphson method is also known as the method of
(A) Tangents (B) Secant
(C) Co-secant (D) Squares

3. The condition of convergence for iterative method of solving a system of simultaneous linear equation is stated that the coefficient matrix should be
(A) Upper triangular matrix (B) Diagonally dominant
(C) Singular matrix (D) Rectangular matrix

4. To find the inverse of matrix A, we can use
(A) Gauss – Elimination method (B) Gauss – Seidal method
(C) Gauss – Jacobi method (D) Newton – Raphson method

5. The relation between E and ∇ is
(A) $\nabla-E^{-1}=1$ (B) $1+\nabla=E^{-1}$
(C) $\nabla=1-E^{-1}$ (D) $\nabla=1+E^{-1}$

6. The (n+1)th and higher orderdifferences of a polynomial of degree 'n' are
(A) Quadratic (B) Parabolic
(C) Linear (D) Zeros

7. The first two terms in the Newton – Gregory forward interpolation formula will give
(A) Linear interpolation (B) Parabolic interpolation
(C) Quadratic interpolation (D) Hyperbolic interpolation

8. When the values of the independent variables are not equally spaced, then we apply
(A) Central difference interpolation formula (B) Newton's forward formula
(C) Newton's backward formula (D) Lagrange's interpolation formula

9. The error in the Trapezoidal rule is the order of
(A) $\frac{h^2}{12}$ (B) $\frac{h^3}{16}$
(C) $\frac{3h}{8}$ (D) h^2

10. Simpson's three-eight rule can be applied only when n is
(A) Odd (B) Even
(C) Prime (D) Multiple of 3

11. Simpson's one-third rule is also called
(A) Parabolic rule (B) Hyperbolic rule
(C) Elliptic rule (D) Trapezoidal rule

12. In deriving Trapezoidal rule, we replace the arc of the curve $y = f(x)$ over each sub-interval by its
(A) Tangent (B) Chord
(C) Diameter (D) Radius

13. Taylor's series method is
(A) Single step method (B) Multi-step method
(C) Iterative method (D) Trial and error method

14. The improved Euler method is based on the averages of
(A) Points (B) Slopes
(C) Curves (D) Both points and slopes

15. How many prior values are required to predict the next value in Milne's methods
(A) 4 (B) 3
(C) 2 (D) 1

16. Adam's corrector formula is

(A) $y_{n+1} = y_n + \frac{h}{24}[55y_{n'} - 59y'_{n-1} + 37y'_{n-2} - 9y'_{n-3}]$

(B) $y_{n+1} = y_n + \frac{h}{24}[55y_{n'} + 59y'_{n-1} - 37y'_{n-2} + 9y'_{n-3}]$

(C) $y_{n+1} = y_n + \frac{h}{24}[9y_{n'+1} + 19y_{n'} - 5y'_{n-1} + y'_{n-2}]$

(D) $y_{n+1} = y_n + \frac{h}{24}[9y_{n'+1} - 19y_{n'} + 5y'_{n-1} + y_{n'-2}]$

17. If $B^2 - 4AC > 0$, then the second order PDE is known as
(A) Elliptic (B) Parabolic
(C) Hyperbolic (D) Laplace equation

18. The nature of the PDE $f_{xx} - 2f_{xy} = 0$ is
(A) Elliptic (B) Parabolic
(C) Hyperbolic (D) Laplace equation

19. The equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f(x, y)$ is also called
(A) Laplace equation (B) Poisson's equation
(C) One-dimensional heat equation (D) Two-dimensional heat equation

20. Bender – Schmidt scheme converges for
(A) λ=1 (B) λ=1/2
(C) λ=3 (D) λ=3/2

PART – B (5 × 4 = 20 Marks)
Answer **ANY FIVE** Questions

21. Find an approximate root of $x\log_{10}^{x} - 1.2 = 0$ by False position method.

22. Find the missing terms in the following tables:

x:	0	1	2	3	4
y:	1	3	9	-	81

23. Prove that $\nabla\Delta = \Delta - \nabla = \delta^2$.

24. Find a third degree polynomial passes through the points $(0,-1), (1,1), (2,1)$ and $(3,-2)$ by using Newton's forward interpolation formula.

25. Compute $f'(0)$ from the following data:

x:	0	1	2	3	4
f(x)	1	2.718	7.381	20.086	54.598

26. Consider the initial value problem $\frac{dy}{dx} = y - x^2 + 1, y(0) = 0.5$ using the modified Euler method. Find $y(0.2)$ with $h = 0.2$.

27. Solve $\frac{\partial^2 u}{\partial x^2} - \frac{2\partial u}{\partial t} = 0$ given $u(0,t) = 0, u(4,t) = 0, u(x,0) = x(4-x)$. Assume h=1, find the values of u upto t = 2 times step.

PART – C (5 × 12 = 60 Marks)
Answer **ALL** Questions

28. a. Fit a straight line and a parabola to the following data and find out which one is most appropriate. Reason out for your conclusion.

x:	0	1	2	3	4
y:	1	1.8	1.3	2.5	6.3

(OR)